

NTIRE 2026: HR Depth from Images of Specular and Transparent Surfaces - Track 2: Metric Mono

Method Description and Technical Report

OVERVIEW

This work presents a two-part system for high-resolution indoor scene understanding: (1) a fine-tuned SAM2 segmentation model for material-aware object segmentation and (2) a metric depth estimation pipeline for the NTIRE 2026 challenge. The central contribution is using the segmentation model to guide depth estimation - SAM2 boundary masks are used to sharpen depth transitions at object edges, particularly at the boundaries of transparent and mirror surfaces where standard depth models fail. Both components are trained on the Booster dataset, allowing the segmentation and depth models to share domain knowledge.

PART 1 - SAM2 SEGMENTATION FINE-TUNING

SAM2 (Segment Anything Model 2) was fine-tuned on the Booster indoor dataset for two mask types - scene geometry masks (mask_00) and clean object masks (mask_cat). The ViT-H image encoder and memory encoder were frozen. Only the mask decoder was fine-tuned using LoRA (rank=4), keeping 97.2% of parameters frozen. This strategy prevents overfitting on the small 228-image training set while adapting the decoder to the specific material categories in the dataset - transparent glass, specular metal, and diffuse surfaces.

SAM2 Training	Value
Base model	SAM2 Large (ViT-H)
Frozen	Image encoder + memory encoder
Trainable	Mask decoder via LoRA rank=4
Trainable params	6.3M / 224M (2.8%)
Training images	188 train / 40 val (scene-level split)
Input resolution	1024×1024
Models trained	Two - mask_00 and mask_cat
Best val IoU	0.8777 (mask_00 model)

SAM2 Config	Value
Epochs	60
Batch size	1
Learning rate	3e-4
LR schedule	5-epoch warmup + cosine annealing
Optimizer	AdamW (weight decay 1e-4)
Mixed precision	FP16 AMP
Grad clip	1.0

Loss Component	Weight
Binary Cross Entropy	0.3
Dice Loss	0.4
Focal Loss ($\alpha=0.8$, $\gamma=2$)	0.2
IoU Loss	0.1

PART 2 - METRIC DEPTH ESTIMATION PIPELINE

Depth Model	Role	Fine-tuned	Weight
DA V2 ViT-L	Primary	Yes - decoder	0.6
ZoeDepth NK	Secondary	No	0.4

Depth Anything V2 ViT-L was fine-tuned on the Booster training set using disparity maps converted to metric depth via stereo calibration. Both stereo cameras (camera_00 and camera_02) were used, doubling effective training data to 376 samples. Only the DPT decoder was trained - the DINOv2 ViT-L encoder remained frozen. ZoeDepth NK served as a complementary ensemble model due to its architecturally distinct bin-based depth estimation, producing uncorrelated errors that reduce RMSE when averaged.

Depth Training	Value
Training data	376 images (camera_00 + camera_02)
Ground truth	disp_00.npy → depth(cm) = f×b/disp×100
Epochs	50 (5-epoch LR warmup + cosine decay)
Optimizer	AdamW (lr=1e-4, wd=1e-4)
Frozen	ViT-L encoder (200M params)
Trainable	DPT decoder (30.9M, 9.2%)
Loss	SILog + 0.1×L1 + 0.1×Gradient
Best val RMSE	5.62 cm (with LSE alignment)

INFERENCE PIPELINE & SAM2 INTEGRATION

Step	Technique
Preprocess	Gray World colour calibration + CLAHE
Multi-scale TTA	3 scales (0.2×, 0.4×, 0.6×) × 4 flips = 12 preds
Ensemble	0.6×DA_V2 + 0.4×ZoeDepth (soft average)
LSE Alignment	Scale+shift regression from training disparity
ToM Inpainting	TELEA inpainting on detected ToM regions
SAM2 Boundary	Depth sharpened at SAM2 mask edges
Smoothing	Bilateral filter on homogeneous regions
Output	float32 .npy, 3008×4112 px, unit: cm

The SAM2 model guides depth post-processing: the fine-tuned mask decoder produces per-pixel segmentation boundaries which are dilated to create edge maps. At detected boundaries, a sharper bilateral filter is applied to preserve depth discontinuities. In homogeneous interior regions, a smoother filter reduces noise. This joint approach addresses the fundamental challenge of transparent and mirror surfaces - the depth model cannot reliably estimate depth at ToM pixels, but SAM2 can accurately locate their boundaries, allowing surrounding context to be inpainted and edges to be sharpened accurately.

Key Value	Detail
DA alignment	scale=0.2897, shift=-317.95 cm
Zoe alignment	scale=0.5828, shift=+11.99 cm
Test images	159 images across 18 scenes

KEY CONTRIBUTIONS & POTENTIAL FOR PUBLICATION

The primary novelty of this work is the use of a fine-tuned segmentation model (SAM2) to guide metric depth estimation at object boundaries - specifically targeting the transparent and mirror surface categories that are evaluated by the competition metric. While multi-model depth ensembles are established, the explicit integration of segmentation masks into depth post-processing for ToM surfaces represents a novel contribution. The combination of LoRA-based SAM2 adaptation, dual-camera training data augmentation, multi-scale TTA ensembling, and segmentation-guided boundary sharpening forms a complete system that achieves competitive results (4th place, Track 2) with significantly fewer trainable parameters than end-to-end approaches. This pipeline could be extended to other challenging surface materials and constitutes a meaningful contribution to the intersection of segmentation and depth estimation research.